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Consumer Valuation of Food Quality and Food Safety Attributes in Vietnam

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Food systems in developing countries are currently undergoing a profound transformation toward high-value products. Appropriate policies are needed to guide this transformation, presupposing good understanding of consumer preferences. We analyze consumers' valuation of different vegetable attributes in metropolitan areas of Vietnam, using contingent valuation techniques and a mediation framework for two specific examples. Consumers are willing to pay an average price premium of 60% for Chinese mustard that is free of chemical residues and of 19% for different convenience attributes of potatoes. Income levels and media have positive impacts on the willingness to pay, partly mediated through consumer perceptions.

Food systems in many developing countries are currently experiencing a profound transformation, which is driven by globalization, urbanization, and rising living standards. This transformation is characterized by the emergence and expansion of modern retailers and integrated supply chains within new institutional setups (Reardon et al.). Food quality and food safety are becoming ever more important, which poses new challenges for supply chain actors to adapt to emerging trends in consumer demand. Products with a bundle of specific attributes are required that are directed to satisfy consumers' needs (Henson and Reardon). Understanding consumers' valuation of these attributes is essential for designing effective policies. While a number of willingness to pay (WTP) and similar valuation studies have been carried out recently (e.g., Florax, Travisi, and Nijkamp; Batte et al.; Poole, Martínez, and Giménez), most of them focus on

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industrialized countries. Since the results cannot simply be transferred, specific research in developing countries is needed, particularly in metropolitan areas, which usually have a leading role in the food system transformation for the rest of the country (Pingali).

We use contingent valuation (CV) methods and household survey data to analyze the determinants and mechanisms of consumer WTP for food quality and food safety attributes in metropolitan areas of Vietnam. A mediation framework (Baron and Kenny) is employed, which was recently applied in an analysis of consumer attitudes toward agrobiotechnology (Moon and Balasubramanian). This approach allows us to get a more functional understanding of processes involved in food attribute valuation. Also, it facilitates the identification of indirect effects (Holmbeck). Thus, valuable information on the role of consumer perceptions is provided. Such information is relevant to policy makers, as perceptions are not solely determined by sociodemographic characteristics, but can be influenced and changed, for instance, through media campaigns.

Given its rapid economic development and recent policy reforms, Vietnam is an interesting developing country to study valuation of food product attributes. Restructuring of food supply chains in Vietnam is observed in the context of ongoing economic liberalization (Maruyama and Trung). New demand patterns are emerging, entailing a growing importance of food quality and food safety at the retail level (Mergenthaler, Weinberger, and Qaim). The horticultural sector is affected in particular (Cadilhon et al.), making it an interesting case for a more thorough investigation. Attributes of fresh horticultural products considered here are food safety in terms of low agrochemical residues and different convenience characteristics.

Food Quality and Food Safety Attributes

In the course of Vietnam's rapid economic development, changing lifestyles and new food consumption patterns are emerging. Especially among the urban middle and upper classes, demand for food quality and food safety is growing. Here, we analyze consumer valuation of a reduction in agrochemical residues and of convenience characteristics as two typical examples of product attributes that are gaining in importance.

Food Safety

As a consequence of the intensification process in agriculture (Jansen et al.; Dung et al.), high intakes of agrochemical residues from consumption of food have been reported in Vietnam (e.g., Kurunthachalam et al.). Due to a number of acute poisonings reported in the mid-1990s, use of some highly toxic pesticides was restricted. Nonetheless, Kim reports that incidents of food-borne diseases are still a problem in Vietnam, even though most are never detected. In 2001, 245 registered outbreaks of food-borne diseases were attributed to microbiological, agrochemical, or natural toxic contamination, and other unknown sources. Fresh vegetables are still the most important source of food risks (Sy, Vien, and Quang). Consumers are particularly worried about chemical residues (Figuié).

Already in 1995, the Vietnamese government launched a "safe vegetables" program (*"rau sach"* or *"rau an toan"*) to counteract food safety problems in horticultural produce. In this context, the Ministry of Agriculture and Rural Development issued a temporary regulation on the production of "safe vegetables," which adopted maximum residue levels from Codex Alimentarius as a benchmark in 1998. Since then, the government has fostered the development of vegetable production with improved food safety. In cooperation with local authorities, the "safe vegetable" label is promoted in annual fairs for farmers and in advertising programs for retailers and consumers.

Within the program, training and technical support is given to farmers to improve management of irrigation water, fertilization, and application of pesticides. Produce is marketed through specialized supply chains in a limited number of "safe vegetable" shops and supermarkets. The municipal health care service is meant to work closely with responsible agencies to control regularly the quality and hygienic conditions in vegetable production and marketing. However, quality controls are mostly organized internally in participating cooperatives. Furthermore, awarding the "safe vegetable" label is not authenticated by a standardized certification process, and no formalized sanction mechanisms exist in case of noncompliance (Moustier et al.). Laboratory analyses are hardly conducted or can detect only few types of pesticides (Tam). The lack of standard enforcement mechanisms involves asymmetric information and distrust between producers and consumers (Hoan, Mergenthaler, and Breisinger). In addition to the "safe vegetables" program, there are also different private standards and labels, for which farmers often cultivate under contract schemes and follow tight technical specifications, or they receive other incentives to secure food safety. So far, fruits and vegetables with formal public or private food safety assurance are still a niche market in Vietnam, including in metropolitan areas. Growth seems to be mainly hampered by lax control, lack of sanction mechanisms, and hence limited consumer trust (Hoang and Nakayasu).

Convenience

While food safety has gained a more prominent role within the food system transformation in Vietnam, convenience in terms of short distances to retail outlets has traditionally been an important aspect for consumers in Vietnam (Cadilhon and Tam). More recently, demand for products with special convenience attributes has also been growing (IFPRI). Convenience often implies preprocessed products, such as vegetables that are already washed and peeled, in order to reduce the time of meal preparation at home. From the supplier side, processing vegetables into convenience foods is interesting, because this can potentially improve market outreach to different consumer segments, particularly in urban areas (Maruyama and Trung). Nevertheless, hardly any research has been carried out so far on quantifying related consumer preferences. We hypothesize that especially high-income consumers in Vietnam are willing to pay extra for convenience attributes.

Direct Valuation of Food Quality and Food Safety Attributes

Contingent Valuation

In CV, WTP approaches draw on stated preferences to estimate the value people place on nonmarket goods. Like analysis based on revealed preferences, WTP analysis is based on utility maximization theory. WTP has traditionally been used for the valuation of public goods (e.g., environmental services). More recently, it has also been applied in market research for private goods (cf. Lusk and Hudson). In our context, WTP can be interpreted as an indicator of demand for vegetables with low agrochemical residues or convenience attributes. Apart from assessing market potentials for specific products, WTP analysis can help to better understand general market trends and identify appropriate policy responses.

Our choice of valuation scenarios was based on several criteria. Elicitation of stated preferences is more reliable for objects that respondents are familiar with. A frequently encountered problem in WTP studies is hypothetical bias, particularly for public goods like environmental services. This is of lesser relevance in our context of marketable private goods. Yet, concrete products and product attributes had to be determined. To value food safety, Chinese mustard (*Brassica rapa* var. *chinensis*)—a close relative of Chinese cabbage—was chosen. This leafy vegetable, which is locally called pakchoi, is an integral component of Vietnamese diets. The edible parts of pakchoi are usually exposed to frequent pesticide applications. Furthermore, due to the vegetable's high perishability, supply for the metropolitan areas mainly comes from peri-urban production that is characterized by very high input intensities (Khai, Ha, and Oborn). This puts the issue of chemical residues high on the agenda. Against this background, we developed a hypothetical scenario in which—through assumed changes in production technology—there would be zero agrochemical residues in the product. Half of the respondents were confronted with a slightly different scenario, in which there was only a partial reduction in agrochemical residues. This was to see if a slight variation in the valuation scenario would affect WTP.

A second scenario was constructed, in which consumers valued convenience attributes of potatoes (*Solanum tuberosum*). Though potatoes are not consumed as widely as leafy vegetables, they are expected to have a bigger potential to be sold with convenience attributes. Respondents were asked for their WTP for fresh potatoes that are washed, peeled, pre-cut, packed, and cooled. For half of them, the cooling attribute was intentionally omitted, with the same intention of slight scenario variation as in the safety scenario.

Price premiums were chosen as payment vehicle within a double bounded discrete choice format to elicit consumers' WTP. This approach draws on the incentive properties of discrete choices and the efficiency of open-ended questions (Haab and McConnell). Based on results of a presurvey, respondents were first confronted with a random incremental price bid between 15% and 100% above the current market price; for the second bid, the range was 0–200%. In order to make price bids more comprehensible to respondents, they were first asked how much they currently pay for pakchoi and potatoes, so that the enumerators could translate the percentage price bids into absolute monetary values. For the analysis, absolute increments (the absolute difference between the price bids and the current market price) were used to represent the premium consumers are, or are not, willing to pay for more food safety and convenience. These transformed first and second bids are used in an interval-censored model. We assume that a consumer has a true WTP of

$$\text{WTP}^* = \beta x + \varepsilon \tag{1}$$

where x is a vector of explanatory variables described further down, β is a vector of coefficients to be estimated, and ε is a normally distributed random error term.

WTP* is not observed, so we have to rely on the range that can be identified from observations in the survey. As explained, two sequential price bids were proposed to respondents. The second bid was higher than the first bid for an initial "yes" response, and lower for a "no" response. The first bid is denoted with P^*, and the second bid with P^H if it was higher, and with P^L if it was lower than P^*. Hence, four possible groups of responses can be observed: ($R1$) respondents answering "yes" to both valuation questions, so that WTP $\geq P^H$; ($R2$) those responding "yes" to the first, and "no" to the second bid, so that $P^* \leq$ WTP $< P^H$; ($R3$) those replying "no" to the first, but "yes" to the second bid, so that $P^L \leq$ WTP $< P^*$; and finally ($R4$) those who said "no" to both sequential bids, so that WTP $< P^L$. This log-likelihood function can be specified to estimate WTP

$$\text{(2)} \quad \ln L = \sum_{R1} \ln\left[1 - \phi\left(\frac{P^H - \beta x}{\sigma}\right)\right] + \sum_{R2} \ln\left[\phi\left(\frac{P^H - \beta x}{\sigma}\right) - \phi\left(\frac{P^* - \beta x}{\sigma}\right)\right] + \sum_{R3} \ln\left[\phi\left(\frac{P^* - \beta x}{\sigma}\right) - \phi\left(\frac{P^L - \beta x}{\sigma}\right)\right] + \sum_{R4} \ln\left[\phi\left(\frac{P^L - \beta x}{\sigma}\right)\right]$$

where ϕ is the standard normal cumulative distribution function. The estimated coefficients β can be directly interpreted as marginal effects. Mean WTP is calculated as

$$\text{(3)} \qquad E(WTP) = \hat{\beta} \cdot \bar{x}.$$

Data

For the empirical analysis, an interview-based survey of households in Vietnam's two big cities, Hanoi and Ho Chi Minh City (HCMC), was conducted between August and October 2005. In total, 499 households were covered in almost all administrative districts of both cities, including urban and peri-urban areas. The sample is a random subsample of the nationally representative Vietnam Living Standard Survey 2002 (VLSS2002) for Hanoi and HCMC. Therefore, the sample does also cover households located in the suburbs and villages surrounding the inner-city districts. These peri-urban districts are characterized as more rural, so the sample represents a broad picture of different types of consumers. Per capita expenditures in peri-urban districts are about 40% lower than average expenditures in the urban districts.

Selected household characteristics used in the estimation of WTP for the food product attributes are shown in table 1. A comparison of some variables of our sample with the full sample of the VLSS2002 for Hanoi and HCMC reveals that households in both samples are very similar. Average annual per capita household expenditures, which we use as a reliable measure of income, amount to around 9.8 million Vietnamese Dong (VND). This corresponds to US$ 615 based on official exchange rates, and US$ 2,960 based on purchasing power parity in 2005. Female labor force participation, often referred to as an important demand-side driver of the food system transformation, is included as a dummy. It takes the value of one

Table 1. Sample statistics

	Variable	Description	Scenario	Own Survey	VLSS2002
Sociodemographics	Education	Years of schooling	Both	8.7 [4.5]	–
		Household head with 12 years of schooling or more	Not included	38.5%	35.3%
	Expenditures	Annual per capita expenditures (million VND)	Both	9.8 [6.4]	9.2 [6.1]
	F-empl	Purchase person is female and employed (dummy)	Both	22.4%	–
	Age	Age of respondent (years)	Both	50.4 [13.4]	–
		Age of household head in years	Not included	52.4 [13.6]	53.6 [14.1]
	HH-size	Household size (heads)	Both	4.8 [7.2]	4.2 [1.9]
	Children	Household with children under 5 (dummy)	Both	23.3%	22.2%
	Urban	Households located in urban districts (dummy)	Both	73.0%	73.8%
	Hanoi	Households located in Hanoi (dummy)	Both	36.7%	36.2%
Media	General	Number of media regularly used	Both	2.0 [0.8]	–
	Specific	Heard or seen specific reports in media (dummy)	Safety	90.2%	–
		Heard or seen advertisements in media (dummy)	Convenience	80.3%	–
Perceptions	Knowledge	Knowledge on pesticide use (dummy)	Safety	75.8%	–
	Concern	Concerns about food safety (dummy)	Safety	93.4%	–
	Trust	Trust in the "safe vegetable" label (dummy)	Safety	33.4%	–
	Price	Price consciousness (dummy)	Both	37.7%	–
	Openness	Openness for new products (dummy)	Convenience	19.6%	–
	Time	Food purchases in the evening (dummy)	Convenience	3.5%	–
Controlling variables	Scenario	Scenario with only partial residue reduction (dummy)	Safety	49.4%	–
		Scenario without cooling (dummy)	Convenience	49.5%	–
	Experience	Previous experience with formal labels (dummy)	Safety	9.6%	–
		Previous experience with convenience (dummy)	Convenience	7.0%	–
	Bid	First price bid (% above market price)	Safety	47.6 [32.9]	–
		First price bid (% above market price)	Convenience	48.0 [33.0]	–
	Price (IV)	Normally paid price for pakchoi, predicted (VND/kg)	Safety	3,877.8 [373.3]	–
		Normally paid price for potatoes, predicted (VND/kg)	Convenience	6,434.5 [1035.9]	–

Notes: The table shows mean values and standard deviations in brackets. Data from the VLSS2002 was adjusted by the official consumer price index for expenditure and by time for age.
[a] "Safety" means the variable was used in the estimation of WTP for food safety and "convenience" in the estimation of WTP for convenience.

if the respondent is female and employed. The educational level of the respondent is measured in years of schooling. Household composition is included with the number of household members and a dummy for the presence of children under the age of five. In addition, the age of the respondent is included. Geographic location is captured by two dummies, one for households located in Hanoi (with HCMC as reference), and the other for households located in urban areas (with peri-urban areas as reference).

Apart from these sociodemographic variables, media are hypothesized to influence WTP of consumers. A general impact of media is captured by the number of media regularly used by the respondent, whereby TV, radio, newspaper and Internet were specifically asked for. On average, respondents regularly use two of these media, with TV and newspaper being the most important ones. For a more specific impact of media, we distinguish between food safety and convenience valuation. Vietnamese media frequently report about cases of negative health impacts (including lethal effects) caused by spoiled and contaminated food. More than 90% of respondents in our sample indicated that they have heard or seen such reports in different media, particularly on TV. This possible impact on WTP is captured by a dummy in the safety valuation. In the convenience model, a different dummy is included that captures whether respondents have heard or seen advertisements for processed horticultural products in the different media.

Consumer perceptions are expected to impact on food safety and quality valuation as well. For safety valuation, knowledge about pesticide use, food safety concerns, trust in the "safe vegetable" certification, and price consciousness are considered. In the case of convenience, we test openness toward new food products, preferred food shopping time in the evening, and price consciousness. These perception variables are represented as dummies,[1] as shown in table 1.

Finally, appropriate controlling variables have to be determined. Since we varied each scenario slightly for half of the respondents, dummies are included to test for the impact of these variations. Furthermore, actual experience with purchases of vegetables with safety and convenience attributes might influence consumer's WTP, so that previous experience dummies are introduced in the models. In spite of their increasing importance, horticultural products with special quality or safety attributes still play a relatively small role in Vietnam. While products with some form of formal safety assurance account for less than 4% of fresh fruit and vegetable markets in metropolitan areas, the share of products with convenience attributes is still lower at around 2%. To take account of a possible starting point bias in the CV format, the first price bid is included in the estimation procedure (for a similar approach cf. Boyle, Bishop, and Welsh). In addition, the price that consumers normally pay for purchased products is added as an independent variable, to take account of unobserved quality preferences.[2]

Direct Impacts on WTP

The estimation results of the WTP models with the above-described independent variables are displayed in table 2 (model 1.1 and 2.1). The dummies for the scenario variations are not statistically significant. A possible explanation is that respondents find it difficult to notice and differentiate between relatively small nuances in scenario descriptions. Therefore, the exact numerical results should be

Table 2. WTP models for food product attributes

	Food Safety			Convenience		
	Model 1.1 Path A&C	**Model 1.2 Path C**	**Model 1.3 Path A**	**Model 2.1 Path A&C**	**Model 2.2 Path C**	**Model 2.3 Path A**
Sociodemographic predictors						
Education	−29.54		0.20	46.11		35.51
	[32.51]		[32.09]	[48.59]		[49.08]
Expenditure	61.97***		69.67***	74.02***		98.67***
	[21.72]		[22.40]	[27.84]		[27.68]
F-empl	−342.37		−422.40	−754.66*		−659.82
	[278.16]		[283.43]	[433.03]		[433.52]
Age	−4.29		−5.45	−27.39*		−31.19**
	[8.90]		[8.98]	[14.12]		[14.20]
HH-size	3.03		−24.15	75.32		71.36
	[57.89]		[57.41]	[86.72]		[87.24]
Children	239.96		431.36	−33.63		17.88
	[269.04]		[273.08]	[420.65]		[425.15]
Urban	508.91*		488.18*	671.1		761.98
	[288.55]		[295.81]	[460.68]		[465.70]
Hanoi	260.05		180.60	−1, 484.27***		−1, 612.38***
	[261.47]		[252.54]	[401.23]		[405.07]
Media predictors						
General	270.29*		382.07***	211.1		403.45*
	[141.47]		[141.07]	[227.24]		[223.45]
Specific	−338.24		9.35	−48.11		−185.03
	[367.28]		[361.58]	[446.51]		[451.24]
Mediators						
Knowledge	671.69**	628.67**				
	[268.60]	[262.03]				
Concerns	1, 632.44***	1, 933.50***				
	[457.98]	[448.77]				
Trust	−151.43	−182.96				
	[238.69]	[227.44]				
Price	−358.51	−501.38**		−842.68**	−1, 235.13***	
	[219.67]	[219.62]		[347.85]	[364.18]	
Openness				996.22**	1, 956.89***	
				[440.86]	[439.63]	
Time				841.54	782.33	
				[894.86]	[963.94]	
Controlling						
Scenario	19.67	−33.04	52.59	−139.03	−194.82	−29.63
	[212.32]	[216.27]	[213.43]	[332.73]	[356.00]	[335.56]
Experience	1, 028.82**	1, 226.38***	988.77**	1, 092.66*	1, 290.30*	1, 094.11*
	[417.42]	[403.92]	[416.26]	[638.41]	[695.64]	[636.65]
First bid	11.95***	12.00***	11.82***	11.76**	10.42*	12.27**
	[3.36]	[3.42]	[3.40]	[5.28]	[5.64]	[5.32]
Price (IV)	0.48	0.84***	0.49	0.15	0.58***	0.15
	[0.31]	[0.26]	[0.31]	[0.18]	[0.16]	[0.18]
Constant	−2, 239.44	−2, 895.66***	−1, 063.17	831.59	−1544.78	370.7
	[1, 380.14]	[1, 117.85]	[1, 328.06]	[1, 442.27]	[1, 077.95]	[1, 444.37]
Log likelihood.:	−541.57	−552.73	−555.22	−694.07	−714.81	−699.92
Chi^2:	96.42	74.09	69.10	102.10	60.61	90.38

Notes: Coefficients can be directly interpreted as marginal effects on WTP (measured in VND/kg), evaluated at sample means. Standard errors are shown in brackets. Results are based on 499 observations.
*, **, *** Significant at the 10%, 5%, and 1% level, respectively.

interpreted with some caution. Previous experience with similar products plays a role in both scenarios, indicating that the markets for these new attributes have been partially exploited already. The first price bid is significant in all models, indicating a starting point bias. We will deal with this issue further down when deriving mean WTP. Normally paid vegetable prices are not significant.

Considering consumer perceptions, general food safety concerns have the highest marginal impact on WTP for zero agrochemical residues: on average, concerned consumers are willing to pay 1,632 VND/kg (around 60%) more than their unconcerned counterparts. The coefficient of knowledge is somewhat lower but also significant. Trust in the "safe vegetable" label and price consciousness do not directly affect WTP for food safety. However, price consciousness significantly reduces WTP for convenience, with a marginal effect of –843 VND/kg (around 50%). Openness, in turn, increases WTP for convenience attributes by 996 VND/kg (around 55%).

Per capita household expenditure is a highly significant determinant of the value consumers attach to food safety and convenience. Each additional million VND of expenditure increases WTP by 62 and 74 VND/kg for the two attributes, respectively. For convenience, age and residence in Hanoi have significantly negative impacts on WTP. Surprisingly, also female labor force participation reduces WTP for convenience. An explanation could be that with the additional income generated and an increase in the opportunity cost of time, meal preparation is delegated to older generation family members rather than used to buy convenience products; this is a typical arrangement of intra-household labor sharing in Vietnamese cities. Urban residence positively influences WTP for food safety, probably reflecting a lack of market transparency. The other sociodemographic variables are not significant in these model specifications.

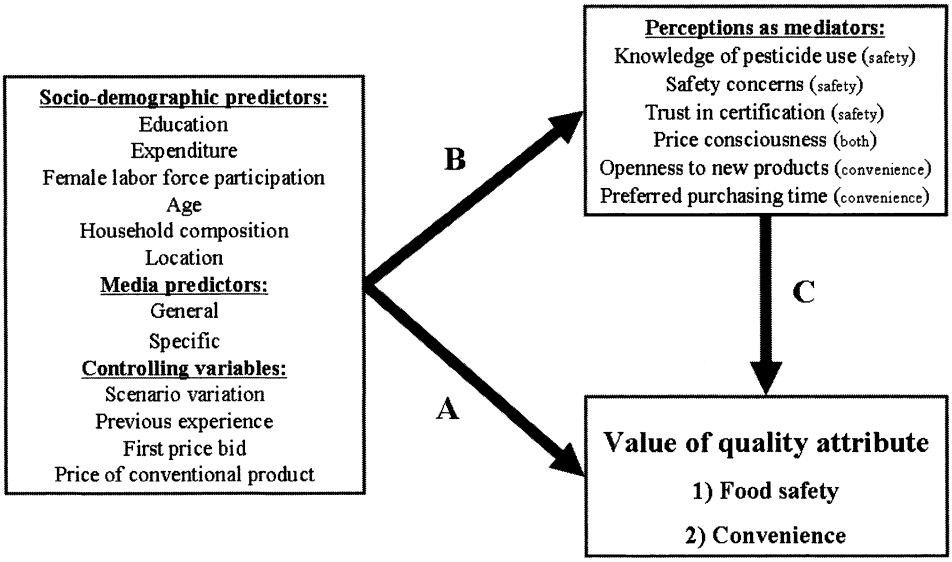
Interesting to observe is the impact of general media use on WTP for food safety but not on convenience. The specific media variable is not significant in both cases: evidently, having heard or seen information specifically related to the attributes valued does not affect WTP. Yet, this variable and some sociodemographic predictors might be associated with indirect effects, which cannot be identified with these particular models. Nor are the mechanisms through which the sociodemographic and media variables affect WTP clear yet. These aspects are analyzed in the following.

Mediation and Indirect Effects through Consumer Perceptions

Mediation Framework

As described, we use consumer perceptions as explanatory variables in addition to sociodemographic and media variables in the WTP models. This is a common approach in empirical valuation studies (e.g., Loureiro, McCluskey, and Mittelhammer). However, there are also studies estimating the impact of sociodemographic variables on consumer perceptions themselves (e.g., Nayga Jr; Knight and Warland). In fact, both of these approaches alone may neglect that consumer perceptions mediate or channel indirect effects of sociodemographic and media variables. The potential to gain a more functional understanding of food attribute

Figure 1. Valuation of food safety and convenience in a mediation framework



Source: Adapted from Baron and Kenny

valuation and the underlying psychological process is omitted when such effects are not considered explicitly. For this reason, we adapt a mediation framework, which was first developed within psychological research (Baron and Kenny) and recently applied in an analysis of consumer attitudes toward agrobiotechnology (Moon and Balasubramanian).

In this framework, we consider consumer perceptions as potential mediators between sociodemographic and media predictor variables and the value consumers attach to improved food quality or safety. Whereas predictors (sociodemographics and media) are external influences that determine the content and way of information processing by the individual, mediators are the result of this information processing. We hypothesize that there is a causal flow from sociodemographic consumer characteristics and media (predictor variables) to consumer perceptions (mediators) and/or to valuation of food attributes (dependent variable), as shown in figure 1. The question we address is whether the predictor variables affect consumer perceptions and food attribute valuation individually or in combination. For example, consumers in households with more media usage might be more knowledgeable and more concerned about pesticides, and in this way, they might value food safety higher. Similarly, older consumers might be less open toward new food products, and in this way they might value convenience attributes less. The main advantage of the mediation framework is that it can detect indirect and mediated effects, which would go unrecognized in a conventional valuation study.

The terms *mediated* and *indirect effects* are at times used interchangeably, though a distinction should better be made. A mediated effect is understood as a special case of an indirect effect, which requires that the direct impact of a predictor on the dependent variable is significant (path A in figure 1). For the assessment of indirect effects, this assumption is not required. It is therefore possible to identify indirect effects, when the direct effect is not significant (Holmbeck). Estimation results for the WTP models with separate specifications for paths A and C (cf. figure 1) are shown in table 2, next to the full models already discussed, which combine paths A and C.

Most mediation studies use a causal steps approach to identify mediation, as first described by Baron and Kenny. This involves comparison of marginal effects across different model specifications. We refrain from doing so, because this method lacks statistical power. The Sobel test is a direct and more statistically rigorous method to test mediation hypotheses; it was also proposed by Baron and Kenny, but has hardly been applied in empirical studies so far (Dearing and Hamilton). The Sobel test is fairly conservative (MacKinnon, Warsi, and Dwyer), so that for small samples bootstrapping has been suggested as a more efficient approach (e.g., Shrout and Bolger). For relatively large samples ($n > 400$), however, the Sobel test is recommended as the best approach (Dearing and Hamilton), so that we employ this method here.

The Sobel test is conducted by assessing the strength of the indirect effect. As in figure 1, the path from the mediators to the dependent variable is denoted as C, the respective marginal effect as c and s_c denotes the corresponding standard error (models 1.2 and 2.2 in table 2). The path from predictors to mediators is denoted as B. For the estimation of these different perception models, we employ probit specifications with a binary representation of consumer perceptions (CP):

$$CP = \gamma z + \mu \tag{4}$$

where z is a vector of explanatory variables consisting of the sociodemographic and media predictors, γ is a vector of coefficients to be estimated, and μ is a random error term. The marginal effect for the respective variable is denoted as b, and the standard error as s_b. Results are displayed in table 3.[3] They are used to calculate indirect effects and to conduct the Sobel test.[4]

Under the assumption of multivariate normality, the indirect effect of b and c can be calculated as

$$x_{bc} = b \cdot c \tag{5}$$

and the standard error of the indirect effect as

$$s_{bc} = \sqrt{c^2 s_b^2 + b^2 s_c^2 + s_c^2 s_b^2}. \tag{6}$$

In order to conduct the Sobel test, the indirect effect x_{bc} is divided by s_{bc} to yield a critical ratio that can be judged against a value from the standard normal distribution appropriate for a given significance level.

Indirect and Mediated Impacts on WTP

Summary results of the magnitude and significance of indirect and mediated effects based on the Sobel test are presented in table 4. For food safety valuation, the effects of sociodemographic variables are not mediated through consumer perceptions, although indirect effects occur for education and Hanoi. For both

Table 3. Probit models to explain consumer perceptions

	Model 1.4 Knowledge	Model 1.5 Concerns	Model 1.6 Trust	Model 1.7 Price	Model 2.4 Openness	Model 2.5 Time	Model 2.6 Price
Sociodemographic predictors							
Education	0.0303***	0.0018	0.0161**	0.0011	−0.0072	−0.0013	0.0003
	[0.0055]	[0.0012]	[0.0067]	[0.0068]	[0.0055]	[0.0020]	[0.0068]
Expenditure	0.0027	0.0012	−0.0019	−0.0080**	0.0112***	0.0008	−0.0081**
	[0.0031]	[0.0010]	[0.0033]	[0.0039]	[0.0027]	[0.0009]	[0.0039]
F-empl	−0.0643	0.0046	−0.1376***	0.0879	0.0792	0.0419	0.0855
	[0.0524]	[0.0068]	[0.0509]	[0.0591]	[0.0505]	[0.0271]	[0.0591]
Age	0.0002	−0.0003	−0.0076***	−0.0009	−0.0044***	−0.0003	−0.0008
	[0.0015]	[0.0003]	[0.0019]	[0.0019]	[0.0016]	[0.0006]	[0.0019]
HH-size	0.0111	−0.0021	0.0092	0.0254**	0.0178*	−0.0025	0.0250**
	[0.0092]	[0.0017]	[0.0115]	[0.0115]	[0.0092]	[0.0040]	[0.0116]
Children	0.019	0.0148*	−0.0212	−0.1043*	−0.0365	−0.0101	−0.1050*
	[0.0430]	[0.0087]	[0.0564]	[0.0535]	[0.0458]	[0.0165]	[0.0536]
Urban	−0.0427	0.0029	0.0229	−0.0082	0.1394***	−0.0218	−0.008
	[0.0408]	[0.0078]	[0.0612]	[0.0583]	[0.0419]	[0.0231]	[0.0583]
Hanoi	−0.1869***	−0.0017	−0.3483***	0.0987*	−0.0837**	0.0168	0.1091**
	[0.0437]	[0.0067]	[0.0436]	[0.0513]	[0.0414]	[0.0174]	[0.0528]
Media predictors							
General	−0.0317	0.0140*	−0.0089	−0.1190***	0.0995***	0.0012	−0.1234***
	[0.0237]	[0.0076]	[0.0302]	[0.0304]	[0.0249]	[0.0094]	[0.0306]
Specific	0.2079***	0.0134	−0.1737**	0.000	−0.1357**	−0.0032	0.0497
	[0.0783]	[0.0158]	[0.0877]	[0.0776]	[0.0642]	[0.0192]	[0.0579]
Log likelihood:	−218.49	−68.82	−274.62	−307.27	−219.29	−67.35	−306.92
Chi^2:	68.13	69.03	95.33	41.48	97.6	10.01	42.19
Pseudo R^2:	0.13	0.33	0.15	0.06	0.18	0.07	0.06

Notes: Coefficients can be directly interpreted as marginal effects on the probabilities to observe individual perceptions, valued at sample means. Standard errors are shown in brackets. Results are based on 499 observations.
*, **, *** Significant at the 10%, 5%, and 1% level, respectively.

Table 4. Sociodemographics' direct and indirect impacts on WTP

	Food Safety					Convenience			
		Indirect through					Indirect through		
	Direct	Knowledge	Concerns	Trust	Price	Direct	Openness	Time	Price
Sociodemographic predictors									
Education	−29.54	19.05**	3.48	−2.95	−0.55	46.11	−14.09	−1.02	−0.37
	[32.51]	[8.78]	[2.52]	[4.15]	[3.73]	[48.59]	[11.48]	[2.78]	[8.76]
Expenditure	61.97***	1.70	2.32	0.35	4.01	74.02***	21.92***	0.63	10.00*
	[21.72]	[2.23]	[2.06]	[1.06]	[2.76]	[27.84]	[7.32]	[1.36]	[5.82]
F-empl	−342.37	−40.42	8.89	25.18	−44.07	−754.66*	154.99	32.78	−105.6
	[278.16]	[39.47]	[13.65]	[34.64]	[37.67]	[433.03]	[107.10]	[52.57]	[82.23]
Age	−4.29	0.13	−0.58	1.39	0.45	−27.39*	−8.61**	−0.23	0.99
	[8.90]	[1.02]	[0.61]	[1.82]	[1.06]	[14.12]	[3.75]	[0.80]	[2.46]
HH-size	3.03	6.98	−4.06	−1.68	−12.74	75.32	34.83*	−1.96	−30.88*
	[57.89]	[6.91]	[3.50]	[3.96]	[8.41]	[86.72]	[20.04]	[5.52]	[17.49]
Children	[239.96]	[12.20]	28.62	3.88	52.29	−33.63	−71.43	−7.9	129.69
	[269.04]	[29.72]	[18.50]	[17.15]	[37.18]	[420.65]	[93.25]	[22.68]	[78.91]
Urban	508.91*	−26.84	5.61	−4.19	4.11	671.10	272.79***	−17.05	9.88
	[288.55]	[29.96]	[15.54]	[18.61]	[31.96]	[460.68]	[104.01]	[35.55]	[75.13]
Hanoi	260.05	−117.50**	−3.29	63.72	−49.49	−1, 484.27***	−163.79*	13.14	−134.75*
	[261.47]	[57.31]	[13.32]	[80.23]	[35.47]	[401.23]	[90.82]	[27.00]	[78.75]
Media predictors									
General	270.29*	−19.93	27.07*	1.63	59.66*	211.10	194.71***	0.94	152.42**
	[141.47]	[18.15]	[16.34]	[9.04]	[30.98]	[227.24]	[66.39]	[11.73]	[59.77]
Specific	−338.24	130.7*	25.91	31.78	0.00	−48.11	−265.55*	−2.50	−61.39
	[367.28]	[76.23]	[31.93]	[47.08]	[42.48]	[446.51]	[141.91]	[24.03]	[76.72]

Notes: Coefficients are marginal effects on WTP (measured in VND/kg), evaluated at sample means. Standard errors are shown in brackets. Results are based on 499 observations.
*, **, *** Significant at the 10%, 5%, and 1% level, respectively, based on the Sobel test.

variables, the indirect effect is through knowledge. Residents of Hanoi are less likely to know about pesticide use in vegetables, which indirectly reduces their WTP for products that are free of agrochemical residues. Better education, in turn, increases knowledge levels and thus also WTP for food safety. The indirect effects of media are also interesting. Consumers who use more media channels on a regular basis are more concerned and less price conscious, thus influencing WTP for food safety positively. Having heard or seen specific media reports increases knowledge, thereby also increasing WTP for residue-free products. Even though having heard or seen media reports about food safety issues undermine trust in the "safe vegetable" label, a significant indirect effect on WTP through the trust variable cannot be established.

For the valuation of convenience, more mediated and indirect effects can be observed. The effect of household expenditure levels on WTP for convenience is positively mediated through higher levels of openness and lower levels of price consciousness. Opposite effects are detected for consumers in Hanoi. Residence in urban districts is positively associated and age is negatively associated with WTP for convenience, in both cases through different levels of openness. Household size has a positive indirect impact through openness, which however is counteracted by a negative indirect effect through price consciousness. The media variables also have different indirect effects on WTP for convenience attributes. General media use positively impacts on WTP through higher levels of openness and lower levels of price consciousness. On the other hand, having heard or seen food advertisements has a negative impact through less openness. The latter effect is somewhat unexpected. A possible explanation is that these advertisements are not appealing to consumers and cause them to be less open toward new food products in general.

Mean WTP

Based on models considering sociodemographic and media predictors (models 1.1 and 2.1 in table 2), we estimate the mean WTP for the food safety attribute to be 74% and for convenience to be 28% higher than the current market price of the respective vegetables (see table 5). Controlling for the starting point bias, mean WTP comes down to 60% and 19% for food safety and convenience, respectively. In the Vietnamese context, it is not surprising that WTP for food safety is higher than for convenience, because average income levels are still relatively low, but awareness of food safety issues is widespread.

How do our estimated values for mean WTP compare with previous results from other studies and countries? In rich countries, mean WTP for products that are free of agrochemical residues is often somewhat lower than what we found here, when comparisons are made in percentage terms relative to the market price of conventional products (cf. Florax, Travisi, and Nijkamp). This should not be surprising, however, because food regulations in poor countries are usually less strict or less strictly enforced. So, the chemicals used are often more toxic and residue levels much higher. The few results that are available for other low- and middle-income countries are more similar to ours. For instance, Fu, Liu, and Hammitt found price increments between 46% and 75% for a leafy vegetable with low pesticide residues in Taiwan. Schmidt and Vanit-Anunchai estimated mean WTP

Table 5. Estimated WTP for food product attributes

	Food Safety		Convenience	
	VND/kg	% of Market Price	VND/kg	% of Market Price
Median WTP	3,028	78%	2,550	40%
Mean WTP	2,885	74%	1,771	28%
Mean less starting point bias	2,316	60%	1,212	19%
Price conscious consumers	2,093	54%	687	11%
Not safety concerned consumers	792	20%		
Price conscious and not safety concerned consumers	569	15%		
Not open consumers			1,016	16%
Price conscious and not open consumers			491	8%

to be almost 100% for "environmentally friendly" produced Chinese cabbage in Thailand, while Krishna and Qaim found a WTP for residue-free vegetables of 57% in India. In Vietnam, certified organic products, of which supply is very limited so far, reaches prices two to three times higher than conventional vegetables in the market (Moustier et al.). At those high price levels, the organic market caters to a very limited segment of Vietnamese consumers. Our results suggest that the range of customers for residue-free vegetables could increase considerably at somewhat lower prices.[5] We are not aware of any previous studies that have estimated WTP for convenience attributes in a developing country context.

The importance of considering consumer perceptions in the valuation of food attributes becomes evident when comparing WTP estimates for different consumer segments. For example, table 5 shows that price consciousness reduces WTP for residue-free products only to a small extent, while the impact of general food safety concerns is relatively large: on average, nonconcerned consumers are willing to pay a much lower premium of only 20%. For convenience, price consciousness has a much more pronounced effect: compared to the average household, price conscious consumers are only willing to pay about half the premium for convenience attributes. Likewise, less openness toward new products reduces WTP for convenience.

Conclusions

Food systems in many developing countries are currently undergoing a profound transformation, with food quality and food safety aspects growing in importance. Against this background, we have analyzed consumers' valuation of food safety and convenience in metropolitan areas of Vietnam. A CV approach has been employed in a mediation framework to estimate consumers' WTP for residue-free Chinese mustard and pre-cut potatoes as specific examples. It was found that sociodemographic and media variables influence WTP indirectly and partly mediated through different consumer perceptions—a finding that deepens our understanding of processes involved in valuation of food attributes. Furthermore, while CV studies have been carried out in several developed countries, only very few studies are available for developing countries, so that our results add to the existing body of literature.

In terms of food safety, we have found that Vietnamese consumers on average are willing to pay 60% more for vegetables that are free of agrochemical residues. In terms of convenience, we have analyzed potential demand for vegetables that are preprocessed so to reduce preparation times within the household, and found a mean WTP of 19%. WTP for both attributes is positively related to household expenditure levels, so that further income growth will fuel demand for safe and convenient food products.

Particular emphasis has been put on the potential role of consumer perceptions as mediators of the effects of sociodemographics and media. It was found, for instance, that education of survey respondents, as well as location variables, could significantly affect consumer perceptions, which leads to indirect effects on WTP for food safety and convenience. In addition, the influence of public media on WTP is mediated or partly channeled indirectly through consumer perceptions. A more widespread and regular use of different media channels is likely to increase consumer demand for new product attributes. While consumer concerns have a large impact on the valuation of food safety, WTP for convenience attributes is mostly channeled through perceptions such as price consciousness and openness. From a policy perspective, public media can and should be used to promote the spread of objective information, especially with respect to health issues.

What are the implications for the food sector as a whole? Supply chains should be ready to respond to the increasing demand for food quality and food safety. Relatively high WTP for new attributes indicates that there is a lot of potential for value added—value that could positively contribute to economic and social development, if captured domestically and shared equitably. This is a challenge for public policy. Convenience characteristics are mostly search attributes, so that markets for convenience products will develop automatically, when living standards rise. For food safety, this might be different, since safety characteristics are credence attributes, which easily lead to market failures. Direct public intervention might be necessary in terms of establishing credible standards and certification systems.

A better understanding of consumer preferences and demand patterns for high-value products is crucial for successfully establishing food policies aimed at efficient and equitable development of supply chains in rapidly changing environments. Therefore, our results should also be of interest to other developing countries experiencing similar trends as Vietnam.

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Endnotes

[1]Consumer perceptions were partly measured on a five point Likert-scale in the survey. Though recoding as dummies leads to a loss of information, the interviews showed that delimitation between the different levels was not always clear-cut among respondents. Use of a threshold level therefore seems justified. Furthermore, we found that due to the distribution of answers, a better split between the different categories of the independent variables was thus achieved. In this approach, we follow Knight and Warland.

[2]An instrumental variable (IV) approach was used to avoid a potential endogeneity problem. Place of purchase dummies were used as instruments to predict the prices of pakchoi and potatoes, which were then included in the WTP estimations.

[3]Note that the magnitude of the marginal effects in tables 2 and 3 cannot be compared directly. While table 3 shows the influence of explanatory variables on the probability to observe certain consumer perceptions, table 2 presents effects on WTP measured in VND/kg.

[4]The specification of paths A, B, and C (figure 1) suggests that the marginal effects for the perception variables shown in table 2 might potentially suffer from an endogeneity bias. This does not affect the accuracy of the Sobel test, however (cf. Dearing and Hamilton).

[5]To add further plausibility to our estimated WTP, it can be compared to total household expenditures. If the average household would switch from conventional pakchoi to pakchoi, which is free of agrochemical residues priced at mean WTP, annual per capita expenditures would increase by 14,120 VND (US$ 0.89), which only constitutes 0.14% of total expenditures. Numbers for the convenience attribute are even lower: switching to conveniently preprocessed potatoes priced at mean WTP would increase per capita expenditure by 7,387 VND (US$ 0.47 or 0.08% of total per capita expenditure).

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